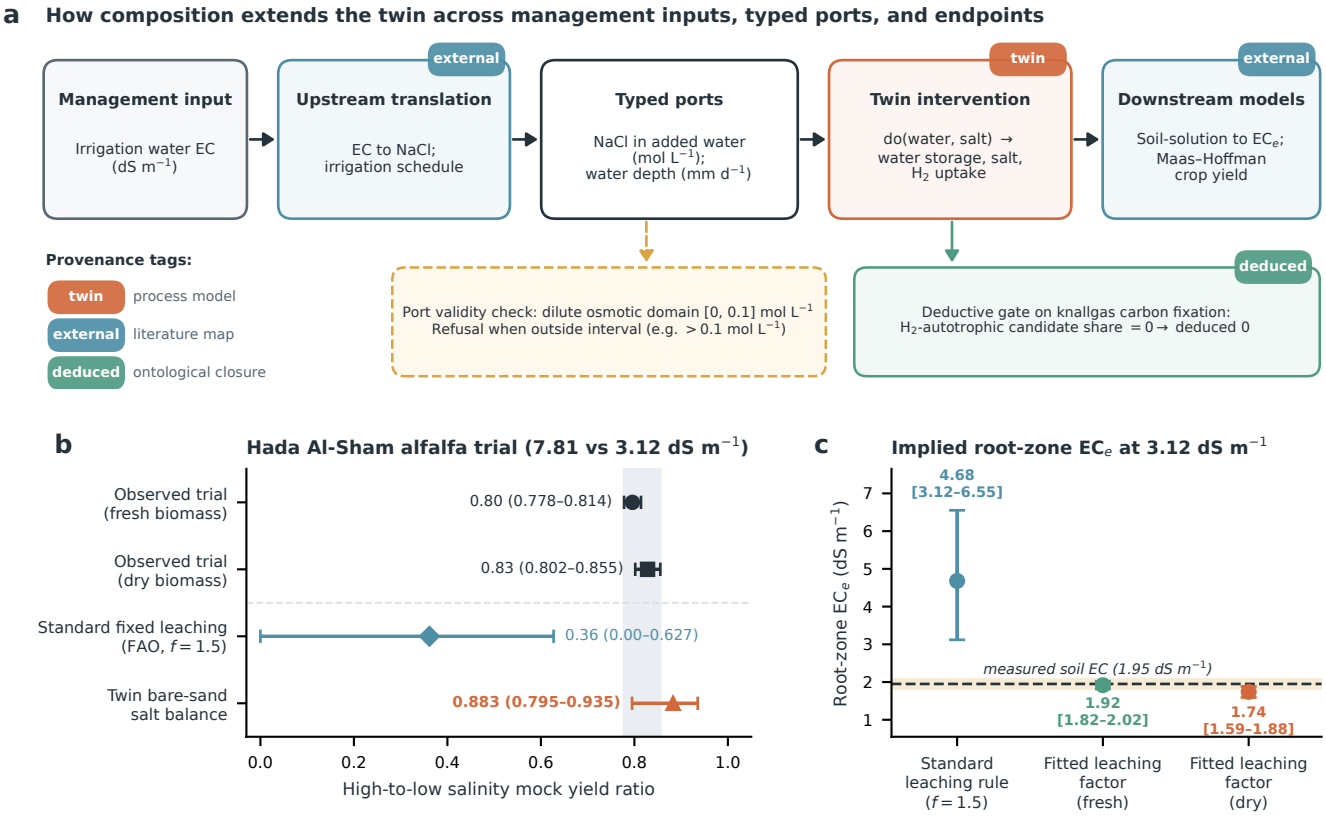
a
How composition extends the twin across management inputs, typed ports, and endpoints
Management input
Irrigation water EC (dS m−1)
external
Upstream translation
EC to NaCl; irrigation schedule
Typed ports
NaCl in added water (mol L−1); water depth (mm d−1)
twin
Twin intervention
do(water, salt) → water storage, salt, H2 uptake
external
Downstream models
Soil-solution to ECe; Maas–Hoffman crop yield
Provenance tags:
twin process model
external literature map
deduced ontological closure
Port validity check: dilute osmotic domain [0, 0.1] mol L−1
Refusal when outside interval (e.g. > 0.1 mol L−1)
deduced
Deductive gate on knallgas carbon fixation:
H2-autotrophic candidate share = 0 → deduced 0
b
Hada Al-Sham alfalfa trial (7.81 vs 3.12 dS m−1)
Observed trial (fresh biomass)
0.80 (0.778–0.814)
Observed trial (dry biomass)
0.83 (0.802–0.855)
Standard fixed leaching (FAO, f = 1.5)
0.36 (0.00–0.627)
Twin bare-sand salt balance
0.883 (0.795–0.935)
0.0
0.2
0.4
0.6
0.8
1.0
High-to-low salinity mock yield ratio
c
Implied root-zone ECe at 3.12 dS m−1
Root-zone ECe (dS m−1)
4.68
[3.12–6.55]
measured soil EC (1.95 dS m−1)
1.92
[1.82–2.02]
1.74
[1.59–1.88]
Standard leaching rule (f = 1.5)
Fitted leaching factor (fresh)
Fitted leaching factor (dry)